

NAISARGI PATEL

🏠 Boston, MA ☎ 857-867-2247 ✉ naisargi56@gmail.com 🌐 Naisargi Patel

EDUCATION

Massachusetts College of Pharmacy and Health Sciences University, Boston, MA 2021 - 2023
Master of Pharmaceutical Economics and Policy-Thesis **GPA: 3.8**
Coursework: Epidemiology, Bio-statistics, Health Service and Outcomes Research, Quantitative methods of analysis, Public Policy and Research, Meta-Analysis, Health Economics and Outcomes Research.

Gujarat Technological University, Gujarat, India 2017 - 2021
Bachelor of Pharmacy- Thesis **GPA: 3.9**
Coursework: Bio-statistics, Pharmacy practice, Regulatory affairs, Pharmacokinetics, Public policy.

WORK EXPERIENCE

Data Science Real-World Evidence Regulatory Researcher June 2023 - Present
Johnson and Johnson - Cambridge, MA

- Applied high-impact **propensity score matched cohort** and **self-controlled case-series analyses** in massive **claims, EHR, inhouse CTs datasets**, uncovering critical **safety** and **efficacy** signals in cancer.
- Employed **machine learning** (random forests, gradient boosting, neural nets) and **NLP** on real-world data for **adverse-event detection** and **treatment-pathway characterization** in cardiac diseases.
- Performed **time-to-event survival analyses** and **mixed-effects models** with **multiple imputation** to assess **longitudinal treatment effects** in neuroscience.
- Developed and validated Cox proportional hazards and LASSO **predictive risk models** for safety endpoints, using **cross-validation** and **calibration plots** across subpopulations in **infectious disease**.
- Constructed **Markov** and **microsimulation health-economic models integrating real-world effectiveness** to estimate **cost-effectiveness** and **budget impact** in immunology.
- Conducted **network meta-analyses** and **Bayesian hierarchical modeling** to synthesize **trial and observational evidence** for **comparative efficacy** and **safety** in cancer.

Research Assistant- Patient reported outcomes and Epidemiology May 2022 - Oct 2022
Brigham and Women's Hospital/ MCPHS University- Boston, MA

- Assisted with Standardization of **ICD-9/10** and CPT coding to classify **geriatric oncology** diagnoses.
- Evaluated clinical outcomes of **inappropriate medication** using **descriptive and inferential analyses**.
- Authored and submitted an **abstract** on **geriatric medication safety** (currently under review).

Health Economics and Evidence Synthesis Co-op May 2022 - Oct 2022
Intas Pharmaceuticals- Ahmedabad

- Performed **cost-utility analysis** integrating **real-world effectiveness** and utilization data to optimize payer strategies for COVID-19 **monoclonal antibody treatments**.
- Developed a **decision-analytic Markov model** using NHDR data to estimate the incremental cost-effectiveness ratio of **monoclonal antibody therapy** in high-risk COVID-19 patients.
- Conducted a **cross-sectional cohort surveillance study** to quantify the **prevalence** and **risk factors** of long COVID symptoms among **recovered COVID-19 patients**.
- Modeled **direct healthcare costs** of **long COVID management** via claims-based cost analyses and **multivariable regression**, informing health-system budgeting and **post-acute care policies**.

SCIENTIFIC MANUSCRIPTS

- Arora, H., Mammi, M., **Patel, N. M.**, Zyfi, D., Dasari, H. R., Yunusa, I., Simjian, T., Smith, T. R., Mekary, R. A. (2023). Dexamethasone and overall survival and progression-free survival in patients with newly diagnosed glioblastoma: A meta-analysis. *Journal of Neuro-Oncology*, 166(1), 17–26. MID: 38151699.
- **Patel, N.** "Pharmacophore Mapping-An Important Tool in Ligand and Structure based drug design, under communication with European Chemical Bulletin(ECB) at Budapest 2025-2026.

POSTER PRESENTATIONS

- 7th Annual Real-World Evidence and Analytics Conference, 2024
 - Presenter: "The Role of Systematic Literature Review and Meta-Analysis in Advancing Clinical Research"
- International Society for Pharmacoeconomics and Outcomes Research (ISPOR) Conference, 2023
 - Presenter: "Comparison of Cardiac Safety Profiles of Metformin and Sulfonylureas in Colorectal Cancer Patients with Type 2 Diabetes Mellitus: A Surveillance, Epidemiology, and End Results (SEER) Medicare Analysis"
- International Society for Pharmacoeconomics and Outcomes Research ISPOR Conference, 2023
 - Presenter: "Assessment of Reporting Adherence for Patient-Reported Outcomes in Aggressive Large B Cell Lymphoma: A Systematic Literature Review of Randomized Controlled Trials in Compliance with CONSORT PRO Extension, CONSORT Outcomes Extension, and ISOQOL Standards"
- International Conference on "COVID-19: Opportunities and Challenges in Pharmaceutical Research, 2021
 - Presenter: "Pharmacophore Modeling in Drug Design"
- International Conference on Molecular Structure Instrumental Approaches, 2020
 - Presenter: "Pharmacophore Mapping - An Important Tool in Ligand and Structure Based Drug Design"

RESEARCH EXPERIENCE

Meta-analysis: Influence of Preoperative use of Dexamethasone on Progression Free or Overall Survival in Patients with Glioblastoma - Brigham and Women's Hospital Boston, MA

- Identified relationships, gaps, contradictions, and inconsistencies in the research.
- Formulated **inclusion and exclusion** criteria, and utilized electronic databases to retrieve papers.
- Extracted and Analyzed data. generated **evidence**, checked the quality by running **I2 statistics**.
- Results: forest plot indicating the non-significant results reported by P-value and confidence intervals.

Cardiac Safety Profile Comparison of Metformin and Sulfonylureas in Colorectal Cancer Patients with Type 2 Diabetes- SEER Medicare Database

- Conducted a retrospective cohort study with seer medicare data (2015-2019) to compare cardiac safety outcomes of metformin vs. sulfonylurea therapy.
- Analyzed major adverse cardiac events (**MACE**) using **kaplan-meier survival** and **cox proportional hazard models**, adjusting for key variables such as age and heart failure history.
- **Primary analysis - statistically significant** factors impacting cardiac risk, and treatment considerations.
- **Secondary analysis** - without covariates adjusting, noted similar **outcomes** and **adverse reactions**.

Cost-Effectiveness Analysis(CEA) of Antibiotics for Respiratory Infections

- Conducted CEA analysis and burden-of-illness studies on antibiotic use for respiratory infections, databases such as **NHS EED, CEA registry, and PubMed** to inform on economic impact.
- Extracted, synthesized, and modeled data in excel, performing **risk-of-bias** assessments to enhance reliability.
- Delivered economic evaluations to support **evidence-based decision-making** in antibiotic interventions, with clear cost-consequence implications for resource allocation.

TECHNICAL SKILLS

- Epidemiology & Outcomes: Cohort, case control, self-controlled case-series; survival and mixed-effects modeling; patient-reported outcomes, machine learning predictive models.
- Health-Economics: CEA, CUA, CBA, CMA; Markov and microsimulation; budget-impact analysis; ICER/QALY.
- Statistical Programming: R, Stata(Basic), SAS(Intermediate), Python (Basic)
- Data & Visualization: SQL/ETL, Excel (pivot/VBA), Shiny/Power BI dashboards, R Markdown reporting.